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Steering actuator apparatus for vehicle

Abstract

A steering actuator apparatus for a vehicle may include: a housing, a driving unit configured to be supported by the housing and generates a driving force, a transmission shaft, which is installed inside the housing to be movable, moves back and forth by receiving the driving force from the driving unit, and varies a steering angle of a wheel, a measurement unit configured to measure the steering angle of the wheel in conjunction with a movement of the transmission shaft, a first rotation prevention unit provided between the housing and the transmission shaft and fixed to the transmission shaft and a second rotation prevention unit, which is detachably coupled to an outside of the housing and interferes with the first rotation prevention unit to prevent relative rotation of the housing and the transmission shaft.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority from and the benefit of Korean Patent Application No. 10-

2022-0003284, filed on Jan. 10, 2022, which is hereby incorporated by reference for all purposes as if set forth herein.

BACKGROUND

Technical Field

(2) Exemplary embodiments of the present disclosure relate to a steering actuator apparatus for a vehicle, and more particularly, to install in a Steer By Wire (SBW).

Discussion of the Background

(3) In general, a Steer By Wire (SBW) system is a steering system that separates a mechanical connection between a steering wheel and a driving wheel of a vehicle, which receives a rotation signal of the steering wheel through an electronic control unit (ECU) and operates a steering motor connected to the driving wheel based on an input rotation signal. The Steer By Wire (SBW) system has advantages such as increasing layout freedom according to a steering system configuration, improving fuel efficiency, and removing disturbances reversely input from the wheels by removing a mechanical connection structure of the existing steering system. The SBW system consists of a steering wheel for a driver to input a steering angle, a reaction device that gives the driver a sense of reaction, and a steering actuator that rotates a tire.

(4) When a rack-driven steering actuator is applied to the SBW system, a sensor is required to detect a steering angle of the wheel through a position of the rack bar. A method of mounting the sensor is divided into a method of attaching the sensor directly to the rack bar and a method of attaching the sensor to a separate device such as a pinion. A linear variable differential transformer (LVDT) sensor shall be applied when the sensor is attached directly to the rack bar. However, in a case of LVDT sensors, since a size is lengthened to detect a range of rack strokes of about 150 mm, there is a problem in which unit price of sensors increases and mass production decreases.

(5) A method of attaching to a separate instrument such as a pinion is to mount an angle sensor instead of a torque sensor on the input pinion side of the existing R-MDPS (EPS). However, the existing input pinion structure is heat-treated on the pinion and rack bar, and there is a problem that cost is excessively consumed due to the large number of assembly parts such as a clearance compensation structure.

(6) The related art of the present disclosure is disclosed in Korean Patent Application No. 10-2009-0116183 (published on Nov. 11, 2009 and entitled “The Support York Structure of The Steering Apparatus for Vehicle”).

SUMMARY

(7) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

(8) An object of the present disclosure is to simplify a coupling structure of a steering angle sensor and provide a steering actuator apparatus for a vehicle capable of stably supporting a rack bar.

(9) In an embodiment, a steering actuator apparatus for a vehicle according to the present disclosure includes: a housing; a driving unit configured to be supported by the housing and generates a driving force; a transmission shaft, which is installed inside the housing to be movable, moves back and forth by receiving the driving force from the driving unit, and varies a steering angle of a wheel; a measurement unit configured to measure the steering angle of the wheel in conjunction with a movement of the transmission shaft; a first rotation prevention unit provided between the housing and the transmission shaft and fixed to the transmission shaft; and a second rotation prevention unit, which is detachably coupled to an outside of the housing and interferes with the first rotation prevention unit to prevent relative rotation of the housing and the transmission shaft.

(10) The transmission shaft may include a ball screw configured to convert the driving force generated from the driving unit into a straight line motion, and a rack bar extending from the ball

screw and one side thereof engaged with the measurement unit.
(11) The first rotation prevention unit may include a first body disposed to surround the other side of the rack bar, a first fixing unit extending from the first body and detachably fixed to an outer surface of the rack bar, and a first rotation prevention member which is concavely recessed from an outer surface of the first body and is coupled to the second rotation prevention unit.
(12) The first body is made of plastic.
(13) The first rotation prevention member is one in which a longitudinal direction thereof extends parallel to a longitudinal direction of the first body.
(14) A plurality of the first rotation prevention members is provided and disposed to be spaced apart from each other at predetermined interval along a circumferential direction of the first body.
(15) The second rotation prevention unit may include a second body in which one side is inserted penetrating an outer surface of the housing, a second rotation prevention member which extends from one side of the second body and is inserted into the first rotation prevention member to limit a rotation of the first body, and a second fixing unit extending from the other side of the second body and detachably fixed to the outer surface of the housing.
(16) An end portion of the second rotation prevention member is formed to be curved with a curvature corresponding to a curvature of the first body.
(17) The second fixing unit may includes a flange extending from the second body and disposed to face a coupling hole provided in the housing, a position adjustment hole formed to penetrate the flange and connects with the coupling hole, and a fastening member which is inserted into the position adjustment hole and fastened to the coupling hole.
(18) A width of the position adjustment hole is greater than a width of the fastening member.
(19) Furthermore, the measurement unit includes a pinion which engages with the rack bar for being combined therewith and rotates in conjunction with a movement of the rack bar, and a sensor coupled to the housing and measuring a rotation angle and a rotation direction of the pinion.
(20) The pinion is formed in a shape of a spur gear.
(21) The pinion is manufactured by an insert injection method.
(22) Furthermore, the driving unit includes a power generator fixed to the housing and generating a rotational force, and a power transmission unit that transmits the rotational force generated by the power generator to the transmission shaft.
(23) The power transmission unit may include a deceleration unit, which is connected to the power generator and amplifies the rotational force generated from the power generator, and a ball nut that is connected to the deceleration unit, and moves the transmission shaft.
(24) The steering actuator apparatus for a vehicle according to the present disclosure may install a measurement unit in a rack bar without a separate component such as a clearance compensation structure or the like, thereby implementing an optimized package and reducing an assembly process.
(25) Furthermore, in the steering actuator apparatus for a vehicle according to the present disclosure, the pinion is provided in the shape of a spur gear and is manufactured by an insert injection method, thereby preventing a division force in an axial direction generated in the pinion and reducing manufacturing costs.
(26) The steering actuator apparatus for a vehicle according to the present disclosure may prevent the transmission shaft from rotating about the central axis inside the housing by the first rotation prevention unit and the second rotation prevention unit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a perspective view schematically illustrating a configuration of a steering actuator

apparatus for a vehicle according to an embodiment of the present disclosure.
(2) FIG. **2** is a perspective view schematically illustrating a configuration except for a housing of the steering actuator apparatus for a vehicle according to an embodiment of the present disclosure.
(3) FIG. **3** is an enlarged perspective view schematically illustrating a configuration of the steering actuator apparatus for a vehicle according to an embodiment of the present disclosure.
(4) FIG. **4** is an enlarged perspective view illustrating the configuration of the steering actuator apparatus for a vehicle according to an embodiment of the present disclosure, which is a different perspective of FIG. **3**.
(5) FIG. **5** is a front view schematically illustrating a configuration of a housing according to an embodiment of the present disclosure.
(6) FIG. **6** is a front view schematically illustrating a configuration of a measurement unit according to an embodiment of the present disclosure.
(7) FIG. **7** is a top view schematically illustrating an installation state of a first rotation prevention unit and a second rotation prevention unit according to an embodiment of the present disclosure.
(8) FIG. **8** is a cross-sectional view schematically illustrating an installation state of the first rotation prevention unit and the second rotation prevention unit according to an embodiment of the present disclosure.
(9) FIGS. **9**A and **9**B are a perspective view schematically illustrating a configuration of the first rotation prevention unit according to an embodiment of the present disclosure.
(10) FIG. **10** is a perspective view schematically illustrating a configuration of the second rotation prevention unit according to an embodiment of the present disclosure.
(11) FIG. **11** is a front view schematically illustrating the configuration of a second fixing unit according to an embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE ILLUSTRATED EMBODIMENTS

(12) The following detailed description is provided to assist the reader in gaining a comprehensive understanding of the methods, apparatuses, and/or systems described herein. However, various changes, modifications, and equivalents of the methods, apparatuses, and/or systems described herein will be apparent after an understanding of the disclosure of this application. For example, the sequences of operations described herein are merely examples, and are not limited to those set forth herein, but may be changed as will be apparent after an understanding of the disclosure of this application, with the exception of operations necessarily occurring in a certain order.
(13) The features described herein may be embodied in different forms and are not to be construed as being limited to the examples described herein. Rather, the examples described herein have been provided merely to illustrate some of the many possible ways of implementing the methods, apparatuses, and/or systems described herein that will be apparent after an understanding of the disclosure of this application.
(14) Advantages and features of the present disclosure and methods of achieving the advantages and features will be clear with reference to embodiments described in detail below together with the accompanying drawings. However, the present disclosure is not limited to the embodiments disclosed herein but will be implemented in various forms. The embodiments of the present disclosure are provided so that the present disclosure is completely disclosed, and a person with ordinary skill in the art can fully understand the scope of the present disclosure. The present disclosure will be defined only by the scope of the appended claims. Meanwhile, the terms used in the present specification are for explaining the embodiments, not for limiting the present disclosure.
(15) Terms, such as first, second, A, B, (a), (b) or the like, may be used herein to describe components. Each of these terminologies is not used to define an essence, order or sequence of a corresponding component but used merely to distinguish the corresponding component from other component(s). For example, a first component may be referred to as a second component, and similarly the second component may also be referred to as the first component.

(16) Throughout the specification, when a component is described as being "connected to," or "coupled to" another component, it may be directly "connected to," or "coupled to" the other component, or there may be one or more other components intervening therebetween. In contrast, when an element is described as being "directly connected to," or "directly coupled to" another element, there can be no other elements intervening therebetween.
(17) The singular forms "a", "an", and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms "comprises/comprising" and/or "includes/including" when used herein, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components and/or groups thereof.
(18) FIG. **1** is a perspective view schematically illustrating a configuration of a steering actuator apparatus for a vehicle according to an embodiment of the present disclosure, FIG. **2** is a perspective view schematically illustrating a configuration except for a housing of the steering actuator apparatus for a vehicle according to an embodiment of the present disclosure, FIG. **3** is an enlarged perspective view schematically illustrating a configuration of the steering actuator apparatus for a vehicle according to an embodiment of the present disclosure, and FIG. **4** is an enlarged perspective view illustrating a configuration of the steering actuator apparatus for a vehicle according to an embodiment of the present disclosure, which is a different perspective of FIG. **3**.
(19) Referring to FIGS. **1** to **4**, a steering actuator aperture for a vehicle **1** according to an embodiment of the present disclosure includes a housing **100**, a driving unit **200**, a transmission shaft **300**, a measurement unit **400**, a first rotation prevention unit **500**, a second rotation prevention unit **600**, and a flow prevention unit **700**.
(20) The housing **100** forms a schematic appearance of the steering actuator apparatus for a vehicle **1** according to an embodiment of the present disclosure and entirely supports the driving unit **200**, the transmission shaft **300**, the measurement unit **400**, the first rotation prevention unit **500**, and the flow prevention unit **700**.
(21) FIG. **5** is a front view schematically illustrating a configuration of a housing according to an embodiment of the present disclosure.
(22) Referring to FIG. **5**, the housing **100** according to an embodiment of the present disclosure may be formed to have a hollow cylindrical shape. The housing **100** is disposed in a longitudinal direction parallel to a width direction of a vehicle. The housing **100** may be fixed to and supported by a sub-frame (not illustrated) installed below the vehicle by bolting or the like. The specific shape of the housing **100** is not limited to the shape illustrated in FIG. **1**, and various changes in design may be made within a technical idea of a shape capable of entirely supporting a configuration of the steering actuator apparatus for a vehicle **1** according to an embodiment of the present disclosure.
(23) The housing **100** is provided with an insertion unit **110**. The insertion unit **110** according to an embodiment of the present disclosure may be formed to have the shape of a hole formed through the outer surface of the housing **100**. The insertion unit **110** is disposed to face the first rotation prevention unit **500**, and the second rotation prevention unit **600** is inserted therein. That is, the insertion unit **110** provides a space in which the first rotation prevention unit **500** and the second rotation prevention unit **600** may pass through the housing **100** and be coupled to each other. A cross-sectional shape and extension length of the insertion unit **110** may be variously changed in design according to a shape of the second rotation prevention unit **600**.
(24) The housing **100** is provided with a coupling hole **120**. The coupling hole **120** according to an embodiment of the present disclosure may be formed to have the shape of a hole formed through the outer surface of the housing **100**. The coupling hole **120** is disposed to be spaced apart from the insertion unit **110** by a predetermined distance in the radial direction of the insertion unit **110**. A

plurality of the coupling holes **120** may be disposed to be spaced apart from each other along the circumferential direction of the insertion unit **110**. On the inner circumferential surface of the coupling hole **120**, a thread extending in a spiral shape along the longitudinal direction of the coupling hole **120** may be provided so as to be screw-coupled with a fastening member **633**.

(25) The driving unit **200** is supported by the housing **100** and generates a driving force to move a transmission shaft **300**, which will be described later, back and forth in a straight line inside the housing **100**.

(26) The driving unit **200** according to an embodiment of the present disclosure includes a power generator **210**, a control unit **220**, and a power transmission unit **230**.

(27) The power generator **210** generates a rotational force by receiving power from the outside. The power generator **210** is fixed to the housing **100** to support overall the control unit **220** and the power transmission unit **230**. The power generator **210** according to an embodiment of the present disclosure may be exemplified as various types of electric motors that convert power received from the outside, such as an AC, a DC BLDC motor, or the like into the rotational force. The power generator **210** may be fixed to an outer circumferential surface of the housing **100** by bolting, welding, or the like.

(28) The control unit **220** is connected to the power generator **210** and controls an operation of the power generator **210** based on a measured value measured by a measurement unit **400** to be described later. The control unit **220** according to an embodiment of the present disclosure is coupled to one side of the power generator **210**, and may be exemplified as various types of electronic control units (ECUs) capable of generally controlling the operation of the power generator **210**, such as a rotation speed and whether or not the power generator **210** is rotated.

(29) The power transmission unit **230** is connected to the power generator **210** and the transmission shaft **300**, and transmits the rotational force generated from the power generator **210** to the transmission shaft **300**.

(30) The power transmission unit **230** according to an embodiment of the present disclosure may include a deceleration unit **231** and a ball nut **232**.

(31) The deceleration unit **231** is connected to the power generator **210** and reduces the rotational speed of the power generator **210** to amplify the rotational force transmitted to the ball nut **232**. The deceleration unit **231** according to an embodiment of the present disclosure may include a driving pulley rotating with an output shaft of the power generator **210**, a driven pulley connected to the ball nut **232**, and a belt installed to surround the drive pulley and the driven pulley, configured to rotate the driven pulley in conjunction with a rotation of the driving pulley.

(32) The ball nut **232** receives a rotational force from the deceleration unit **231** and moves the transmission shaft **300**. The ball nut **232** according to an embodiment of the present disclosure may be formed to have the shape of a hollow ring disposed to surround an outer circumferential surface of the transmission shaft **300**. The ball nut **232** is provided with a thread on the inner circumferential surface and is coupled to a ball screw **310** provided in the transmission shaft **300** through a motor such as a ball. When the power generator **210** is driven, the ball nut **232** is rotated about a central axis of the transmission shaft **300** together with the driven pulley.

(33) The specific shape of the power transmission unit **230** is not limited to the shape illustrated in FIGS. **1** and **2**, and various change in design may be made within the technical idea capable of transmitting the rotational force generated from the power generator **210** to the transmission shaft **300**.

(34) The transmission shaft **300** is installed inside the housing **100** to be movable in a direction parallel to the longitudinal direction of the housing **100**. Both ends of the transmission shaft **300** are connected to a pair of tie rods (not illustrated) connected to the wheels of a vehicle, respectively. The transmission shaft **300** receives a driving force from the driving unit **200** and reciprocates inside the housing **100**. The transmission shaft **300** varies an angle of the wheel by transmitting a force to a tie rod connected to both ends thereof as the transmission shaft **300** reciprocates inside

the housing **100**.

(35) The transmission shaft **300** according to an embodiment of the present disclosure includes a ball screw **310** and a rack bar **320**.

(36) The ball screw **310** forms an outer appearance of one side of the transmission shaft **300**. The ball screw **310** is connected to the driving unit **200** and converts a rotational force generated from the driving unit **200** into a straight motion. Accordingly, the ball screw **310** may induce the transmission shaft **300** to reciprocate in a straight line inside the housing **100**. The ball screw **310** according to an embodiment of the present disclosure may be formed to have the shape of a rod having a thread provided on an outer circumferential surface thereof. The ball screw **310** is disposed in a longitudinal direction parallel to the longitudinal direction of the housing **100**. An outer circumferential surface of the ball screw **310** is coupled to an inner circumferential surface of the ball nut **232** through a rolling element such as a ball.

(37) The rack bar **320** extends from the ball screw **310** to form the other side appearance of the transmission shaft **300**. The rack bar **320** according to an embodiment of the present disclosure may be formed to have the shape of a rod extending from one end (right end with reference to FIG. **2**) of the ball screw **310** in a direction parallel to the longitudinal direction of the ball screw **310**. The rack bar **320** may be integrally manufactured with the ball screw **310**, may be manufactured as a separate product from the ball screw **310**, and coupled to the ball screw **310**. A rack gear **321** is formed on one side surface of the rack bar **320** in a longitudinal direction of the rack bar **320**. The rack gear **321** may be formed in the shape of a steel gear that does not heat treatment. The rack gear **321** is engaged with a pinion **410** for being combined therewith provided in the measurement unit **400**.

(38) The measurement unit **400** measures a steering angle of the wheel in conjunction with a movement of the transmission shaft **300**. More specifically, the measurement unit **400** indirectly measures the steering angle of the wheel using a moving direction and a moving distance inside the housing **100** of the transmission shaft **300**. The measurement unit **400** is connected to the control unit **220** by a wirelessly or wired manner to transmit a measured data to the control unit **220**.

(39) FIG. **6** is a front view schematically illustrating a configuration of a measurement unit according to an embodiment of the present disclosure.

(40) Referring to FIGS. **1** to **6**, the measurement unit **400** according to an embodiment of the present disclosure includes a pinion **410** and a sensor **420**.

(41) The pinion **410** is engaged with the rack bar **320** for being combined therewith and rotates in conjunction with a movement of the rack bar **320**. A central axis of the pinion **410** according to an embodiment of the present disclosure is disposed in a direction perpendicular to a longitudinal direction of the transmission shaft **300**. An outer circumferential surface the pinion **410** is engaged with the rack gear **321** for being combined therewith. The pinion **410** rotates clockwise or counterclockwise about the central axis by the rack gear **321** engaged for being combined therewith when the transmission shaft **300** moves.

(42) In the case of a SBW system, a mechanical connection between a steering shaft and the transmission shaft **300** is not required, so that the pinion **410** may be provided in the shape of a spur, that is, a flat gear. Accordingly, when the pinion **410** rotates in conjunction with the rack gear **321**, it is possible to prevent a generation of division force in the pinion **410** in the axial direction, so that the pinion **410** may be made of a plastic material with relatively low rigidity compared to the previous one. In this case, the pinion **410** may be manufactured by an insert injection method without additional gear teeth processing. Accordingly, the pinion **410** may be easily manufactured and reduce manufacturing costs due to conventional heat treatment processing or the like.

(43) The sensor **420** is coupled to the housing **100** and rotatably supports the pinion **410**. The sensor **420** is provided to measure a rotation angle and a rotation direction of the pinion **410** rotated by the rack bar **320**. The sensor **420** is connected to the control unit **220** by a wirelessly or wired manner to transmit the measured data to the control unit **220**. The sensor **420** according to an

embodiment of the present disclosure may include a case that is fixed to the outer circumferential surface of the housing **100** by bolting, welding, or the like to rotatably support the pinion **410**, and various types of rotation angle sensors built into the case to measure the pinion **410**.

(44) A first rotation prevention unit **500** is provided between the housing **100** and the transmission shaft **300** and is fixed to the transmission shaft **300**. A second rotation prevention unit **600** is detachably coupled to the outside of the housing **100** and interferes with the first rotation prevention unit **500** to prevent relative rotation of the housing **100** and the transmission shaft **300**. More specifically, the first rotation prevention unit **500** and the second rotation prevention unit **600** are disposed inside and outside the housing **100** respectively with respect to the housing **100**, so that the transmission shaft **300** is provided to allow the transmission shaft **300** to reciprocate in the longitudinal direction of the housing **100** and to prevent the transmission shaft **300** from rotating around the central axis inside the housing **100**. Accordingly, for the pinion **410** and the rack gear **321** are provided in the shape of a spur-type gear, during steering operation, it is possible to prevent the transmission shaft **300** from rotating around the central axis inside the housing **100** by the rotational moment from the axle shaft or the like.

(45) FIG. **7** is a top view schematically illustrating an installation state of a first rotation prevention unit and a second rotation prevention unit according to an embodiment of the present disclosure, FIG. **8** is a cross-sectional view schematically illustrating an installation state of the first rotation prevention unit and the second rotation prevention unit according to an embodiment of the present disclosure, FIGS. **9**A and **9**B are a perspective view schematically illustrating a configuration of the first rotation prevention unit according to an embodiment of the present disclosure, and FIG. **10** is a perspective view schematically illustrating a configuration of the second rotation prevention unit according to an embodiment of the present disclosure.

(46) Referring to FIGS. **7** to **10**, the first rotation prevention unit **500** according to an embodiment of the present disclosure includes a first body **510**, a first fixing unit **520**, and a first rotation prevention member **530**.

(47) The first body **510** forms a schematic appearance of the first rotation prevention unit **500** according to an embodiment of the present disclosure, and entirely supports the first fixing unit **520** and the first rotation prevention member **530**. The first body **510** according to an embodiment of the present disclosure may be formed to have the shape of a hollow cylinder one side opened. The first body **510** is disposed to surround the circumference of the other side of the rack bar **320**, that is, the circumference surface of the rack bar **320** without the rack gear **321**. The opened side of the first body **510** is disposed to face the rack gear **321** to expose the rack gear **321** to the outside. The first body **510** is integrally coupled to the rack bar **320** through the first fixing unit **520** and moves back and forth in a straight line with the rack bar **320**. The first body **510** may be disposed to be spaced apart from the inner surface of the housing **100** by a predetermined distance so that the outer surface thereof does not interfere with the housing **100** when the rack bar **320** moves back and forth in a straight line. The first body **510** may be made of a plastic material that is light in weight and easy to manufacture and process.

(48) The first fixing unit **520** extends from the first body **510** and is detachably fixed to the outer surface of the rack bar **320**. Accordingly, the first fixing unit **520** may prevent the first body **510** from rotating relative to the rack bar **320** or moving relative to the longitudinal direction of the rack bar **320**. The first fixing unit **520** according to an embodiment of the present disclosure includes a hook unit **521** protruding from both ends of the first body **510** in a ring shape and caught and coupled to the outer surface of the rack bar **320** and a bolting unit **522** extending from the center of the first body **510** in the circumferential direction of the first body **510** and coupled by bolting to the outer surface of the rack bar **320**.

(49) The first rotation prevention member **530** may be formed to have the shape of a groove that is concavely recessed from the outer surface of the first body **510**. The first rotation prevention member **530** is caught and coupled to the second rotation prevention unit **600** to prevent the

relative rotation of the first body **510** with respect to the housing **100**. The first rotation prevention member **530** according to an embodiment of the present disclosure is formed to have a substantially rectangular cross-section and extends in a longitudinal direction parallel to the longitudinal direction of the first body **510**. Accordingly, the first rotation prevention member **530** may limit the rotation of the first body **510** in the circumferential direction of the housing **100** and allow the first body **510** to move back and forth in a straight line along the longitudinal direction of the housing **100** when the first rotation prevention member **530** is caught and coupled to the second rotation prevention unit **600**. A plurality of the first rotation prevention members **530** may be provided and disposed to be spaced apart from each other at predetermined interval along the circumferential direction of the first body **510**. Although FIG. **6** illustrates that a pair of the first rotation prevention members **530** are provided as an example, the number of the first rotation prevention members **530** is not limited to these matters, and various changes in design may be made into three, four, or the like.

(50) Referring to FIGS. **7** to **10**, the second rotation prevention unit **600** according to an embodiment of the present disclosure includes a second body **610**, a second rotation prevention member **620**, and a second fixing unit **630**.

(51) The second body **610** forms the central appearance of the second rotation prevention unit **600** according to an embodiment of the present disclosure, and entirely supports the second rotation prevention member **620** and the second fixing unit **630**. The second body **610** according to an embodiment of the present disclosure is formed to have a substantially cylindrical shape, and one side thereof (left side with reference to FIG. **8**) passes through the insertion unit **110** and is inserted through the outer surface of the housing **100**. A diameter of the second body **610** may be formed to be smaller than a diameter of the insertion unit **110**.

(52) The second rotation prevention member **620** extends from one side of the second body **610** and is inserted into the first rotation prevention member **530** to limit the rotation of the first body **510**. The second rotation prevention member **620** according to an embodiment of the present disclosure may be formed to have the shape of a protrusion protruding from one end of the second body **610** inserted through the insertion unit **110** toward the first body **510**. The end of the second rotation prevention member **620** is inserted into the first rotation prevention member **530** concavely recessed from the outer surface of the first body **510** and is caught and coupled to the first rotation prevention member **530**. In this case, the inner surface of the end portion of the second rotation prevention member **620** may be formed to be curved with a curvature corresponding to a curvature of the first body **510**. Accordingly, the inner surface of the end portion of the second rotation prevention member **620** may stably adhere to the bottom surface of the first rotation prevention member **530** to prevent separation from the first rotation prevention member **530**. A plurality of the second rotation prevention members **620** may be provided and individually be inserted into each of the first rotation prevention members **530**. The number and protruding angle of the second rotation prevention member **620** may be variously changed in design depending on the number and arrangement state of the first rotation prevention member **530**.

(53) The second fixing unit **630** extends from the other side of the second body **610** and is detachably fixed to the outer surface of the housing **100** to support the second body **610** with respect to the housing **100**.

(54) FIG. **11** is a front view schematically illustrating the configuration of a second fixing unit according to an embodiment of the present disclosure.

(55) Referring to FIGS. **6** to **11**, the second fixing unit **630** according to an embodiment of the present disclosure includes a flange **631**, a position adjustment hole **632**, and a fastening member **633**.

(56) The flange **631** extends from the second body **610** and is disposed to face the coupling hole **120** provided in the housing **100**. The flange **631** according to an embodiment of the present disclosure may be formed to have the shape of a plate extending in a radial direction of the second

body **610** from the other end portion (right side with reference to FIG. **8**). A plurality of the flanges **631** may be provided and disposed to be spaced apart from each other in the circumferential direction of the second body **610**. In this case, the flange **631** may be formed in a number corresponding to a plurality of the coupling holes **120**. A plurality of the flanges **631** are disposed such that the inner side thereof faces each of the coupling holes **120**.
(57) The position adjustment hole **632** is formed to penetrate the flange **631** and connects with the coupling hole **120**. The position adjustment hole **632** according to an embodiment of the present disclosure may be formed to have the shape of a hole vertically penetrating the outer surface and the inner surface of the flange **631**.
(58) The fastening member **633** is inserted into the position adjustment hole **632** and fastened to the coupling hole **120**. The fastening member **633** according to an embodiment of the present disclosure may be formed to have the shape of a rod having a circular cross section. One end portion of the fastening member **633** penetrates the position adjustment hole **632** and is inserted into the coupling hole **120**. The fastening member **633** may be provided with a thread extending in a spiral shape along the longitudinal direction of the fastening member **633** on the outer circumferential surface so as to be screwed with the coupling hole **120**.
(59) A head portion **633***a* extending in the radial direction of the fastening member **633** may be provided at the other end portion of the fastening member **633**. A width of the head portion **633***a* is formed to be greater than one of the left and right sides or one of the upper and lower sides of the position adjustment hole **632**. The head portion **633***a* is disposed such that an inner side surface thereof faces the flange **631**. As the fastening member **633** is inserted into the position adjustment hole **632** by a predetermined distance or more, the inner surface of the head portion **633***a* is in close contact with the flange **631** to press and fix the flange **631** to the housing **100**.
(60) A width of the position adjustment hole **632** may be formed to be greater than a width of the fastening member **633**. The position adjustment hole **632** according to an embodiment of the present disclosure may be formed in the shape of a long hole with a substantially elliptical cross section. In this case, the vertical and horizontal widths of the position adjustment hole **632** are formed to be larger than the width of the fastening member **633**. Either the vertical width or the horizontal width of the position adjustment holes **632** is formed to be smaller than the width of the head portion **633***a*. Accordingly, when the fastening member **633** and the coupling hole **120** are temporarily assembled, that is, when the head portion **633***a* is spaced apart from the flange **631**, a position of the second body **610** is adjusted within the range between the position adjustment hole **632** and the fastening member **633**, and the second body **610** may prevent the second rotation prevention member **620** from being inserted in a state that deviates from the first rotation prevention member **530** and induce smooth slide movement of the first body **510**.
(61) Hereinafter, an assembly process of the first rotation prevention unit **500** and the second rotation prevention unit **600** according to an embodiment of the present disclosure will be described.
(62) First, the first body **510** fixed to the outer surface of the rack bar **320** through the first fixing unit **520** is disposed inside the housing **100**.
(63) Thereafter, one side of the second body **610** is inserted into the insertion unit **110**.
(64) In the process of inserting one side of the second body **610** into the insertion unit **110**, the second rotation prevention member **620** is inserted deviating from the first rotation prevention member **530**, and a jamming phenomenon between the second rotation prevention member **620** and the first rotation prevention member **530** may occur.
(65) In order to solve this jamming phenomenon, the fastening member **633** is temporarily assembled to the coupling hole **120** and the position of the second body **610** is adjusted.
(66) More specifically, as the fastening member **633** is not completely inserted into the coupling hole **120**, the inner surface of the head portion **633***a* is spaced apart from the outer surface of the flange **631**. Accordingly, the second body **610** may be moved left and right, up and down within a

range of a gap between the position adjustment hole **632** and the fastening member **633**, or may be rotated about the central axis thereof, and the coupling position with respect to the housing **100** may be adjusted.
(67) Then, the second body **610** moves the rack bar **320** left and right along the longitudinal direction of the housing **100** and correctly mounts the second rotation prevention member **620** on the first rotation prevention member **530**.
(68) Thereafter, the fastening member **633** is completely inserted into the coupling hole **120** to fix the flange **631** to the housing **100**.
(69) A flow prevention unit **700** supports the transmission shaft **300** inside the housing **100**. More specifically, the flow prevention unit **700** prevents the transmission shaft **300** from moving out of a normal position and flowing in the radial direction of the housing **100** inside the housing **100**. Accordingly, the flow prevention unit **700** may prevent the transmission shaft **300** from being jammed or colliding with an inner wall of the housing **100** due to deflection or the like. The flow prevention unit **700** according to an embodiment of the present disclosure is formed to have the shape of a hollow ring and is inserted between the inner circumferential surface of the housing **100** and the outer circumferential surface of the transmission shaft **300**. The flow prevention unit **700** is disposed on one side (right side with reference to FIG. **3**) with respect to the measurement unit **400**. The outer circumferential surface of the flow prevention unit **700** pressed into and fixed to the inner circumferential surface of the housing **100**. The inner circumferential surface of the flow prevention unit **700** is in contact with the transmission shaft **300**, and more specifically, the outer circumferential surface of the rack bar **320** on which the rack gear **321** is formed. The inner circumferential surface of the flow prevention unit **700** may be made of a material having a low friction coefficient so as to stably support the transmission shaft **300** and not to interfere excessively with the movement of the transmission shaft **300**.
(70) Although exemplary embodiments of the present disclosure have been disclosed for illustrative purposes, those skilled in the art will appreciate that various modifications, additions and substitutions are possible, without departing from the scope and spirit of the disclosure as defined in the accompanying claims. Thus, the true technical scope of the present disclosure should be defined by the following claims.

Claims

1. A steering actuator apparatus for a vehicle, comprising; a housing; a driving unit configured to be supported by the housing and generate a driving force; a transmission shaft, installed inside the housing, adapted to move back and forth by receiving the driving force from the driving unit, and vary a steering angle of a wheel; a measurement unit configured to measure the steering angle of the wheel in conjunction with a movement of the transmission shaft; a first rotation prevention unit provided between the housing and the transmission shaft and fixed to the transmission shaft; and a second rotation prevention unit detachably coupled to an outside of the housing and interferes with the first rotation prevention unit to prevent relative rotation of the housing and the transmission shaft.
2. The steering actuator apparatus of claim 1, wherein the transmission shaft comprises: a ball screw configured to convert the driving force generated from the driving unit into a straight line motion; and a rack bar extending from the ball screw and having one side thereof engaged with the measurement unit.
3. The steering actuator apparatus of claim 2, wherein the first rotation prevention unit comprises: a first body disposed to surround another side of the rack bar; a first fixing unit extending from the first body and being detachably fixed to an outer surface of the rack bar; and a first rotation prevention member concavely recessed from an outer surface of the first body and being coupled to the second rotation prevention unit.

4. The steering actuator apparatus of claim 3, wherein the first body is made of plastic.
5. The steering actuator apparatus of claim 3, wherein the first rotation prevention member has a longitudinal direction thereof that extends parallel to a longitudinal direction of the first body.
6. The steering actuator apparatus of claim 5, wherein a plurality of the first rotation prevention members is provided and disposed to be spaced apart from each other at predetermined intervals along a circumferential direction of the first body.
7. The steering actuator apparatus of claim 3, wherein the second rotation prevention unit comprises: a second body in which one side is inserted penetrating an outer surface of the housing; a second rotation prevention member which extends from one side of the second body and is inserted into the first rotation prevention member to limit a rotation of the first body; and a second fixing unit extending from the other side of the second body and detachably fixed to the outer surface of the housing.
8. The steering actuator apparatus of claim 7, wherein an end portion of the second rotation prevention member is formed to be curved with a curvature corresponding to a curvature of the first body.
9. The steering actuator apparatus of claim 7, wherein the second fixing unit comprises: a flange extending from the second body and disposed to face a coupling hole provided in the housing; a position adjustment hole formed to penetrate the flange and connect with the coupling hole; and a fastening member inserted into the position adjustment hole and fastened to the coupling hole.
10. The steering actuator apparatus of claim 9, wherein a width of the position adjustment hole is greater than a width of the fastening member.
11. The steering actuator apparatus of claim 2, wherein the measurement unit comprises: a pinion that engages with the rack bar and rotates in conjunction with a movement of the rack bar; and a sensor coupled to the housing adapted to measure a rotation angle and a rotation direction of the pinion.
12. The steering actuator apparatus of claim 11, wherein the pinion is formed in a shape of a spur gear.
13. The steering actuator apparatus of claim 12, wherein the pinion is manufactured by an insert injection method.
14. The steering actuator apparatus of claim 1, wherein the driving unit comprises: a power generator fixed to the housing and adapted to generate a rotational force; and a power transmission unit adapted to transmit the rotational force generated by the power generator to the transmission shaft.
15. The steering actuator apparatus of claim 14, wherein the power transmission unit comprises: a deceleration unit connected to the power generator and adapted to amplify the rotational force generated from the power generator; and a ball nut that is connected to the deceleration unit, and adapted to move the transmission shaft.